CHAPTER 13

Parametric representations and Cartesian equations

S65, S67, S70

> **Key point · Objectives.** Describe a line and a plane; determine intersections; choose a method for a projection.

Course notes

Throughout this chapter, coordinates are given in an orthonormal frame of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \qquad (t \in \mathbb{R}).$$

A point belongs to the line if there exists *one and the same* real number t satisfying the three equalities. A line has many representations: change the starting point or multiply the direction vector by a non-zero real number.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Cartesian planes

> **Key point · Equation of a plane.** If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a, b, c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, hence belongs to the plane.

Example. The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x - 1) - (y - 2) + 3z = 0$, that is $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single intersection point; an identity gives a line contained in the plane; a contradiction gives a parallel line that is disjoint from the plane. For two planes, non-collinear normals give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normals are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use independent parameters s, t and solve the three equations. With no solution and non-collinear direction vectors, the lines are not coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, look for the orthogonal projection of M onto the normal line through M: $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learned in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Essential check. After calculating a projection, check simultaneously that $H \in \mathcal{P}$ and that $\overrightarrow{MH}$ is collinear with the normal.

Research activity

Three lines facing a plane — 30 min.

In an orthonormal frame of space, consider the plane $\mathcal{P}$ with equation $x + y + z = 3$. In each question, the parameter t ranges over $\mathbb{R}$.

1. The line d_1 has the representation $(x, y, z) = (t, t, t)$. Determine $d_1 \cap \mathcal{P}$.
2. The line d_2 has the representation $(x, y, z) = (3 + t, -t, 0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line d_3 has the representation $(x, y, z) = (t, -t, 0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, state whether the equation obtained after substitution into that of the plane has a unique solution, no solution, or every real number as a solution. Link these cases to the geometric positions found.

Progressive exercises

E1 · Practice. In a frame of space, consider $A(1,-1,2)$ and $B(3,0,1)$. Give a parametric representation of the line (AB).

E2 · Practice. In an orthonormal frame, determine a Cartesian equation of the plane through $A(1,2,3)$ with normal vector $\vec{n} = (1,0,-2)$.

E3 · Practice. In a frame of space, the line d is given by $(x,y,z) = (t, 1+t, 2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x + 2y + z = 8$. Determine their intersection.

E4 · Practice. In a frame of space, determine the intersection of the planes with equations $x + y + z = 1$ and $x - y = 0$. Give a parametric representation if the intersection is a line.

E5 · Practice. In a frame of space, study the relative position of the planes with equations $2x - y + z = 3$ and $4x - 2y + 2z = 5$. Justify.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5–6). At home after S70. Justify each answer; for a false statement, explain precisely what refutes it. The paper is available here: official past paper. Use the data and instructions of this past paper.

Homework

Homework 12 — Projection and intersection — 55 min.

In an orthonormal frame of space, consider the point $M(3,0,0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a non-zero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection H of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. Does the point H belong to this line? Justify.
5. Determine the coordinates of the point K of this line whose third coordinate (the "cote") is zero. Calculate MK, then compare MK and MH.

Teaching notes and solutions

> **Teacher note · Teacher notes.** Use two different parameters to compare two lines. Check all three coordinates, even if two suffice to calculate candidates. A Cartesian equation on its own generally describes a plane in space; it does not describe a line. Reuse the projection from chapter 9 to give meaning to the solving of systems.

Solutions to the activity

1. A point of d_1 has coordinates (t, t, t). It belongs to the plane if and only if $3t = 3$, that is $t = 1$. The intersection is therefore the single point $(1, 1, 1)$.
2. For a point of d_2, $x + y + z = 3 + t - t + 0 = 3$ for every t. All points of the line are in the plane: $d_2 \subset \mathcal{P}$.
3. For a point of d_3, $x + y + z = t - t + 0 = 0$. The equation $0 = 3$ has no solution: the line is disjoint from the plane. Its direction vector $(1, -1, 0)$ is orthogonal to the normal $(1, 1, 1)$; it is parallel to the plane.
4. After substitution, we obtain respectively an equation with a single solution, an identity true for every parameter value, and a contradiction. These three cases correspond to a single intersection point, containment, and no intersection.

E1. $\overrightarrow{AB} = (2, 1, -1) \neq \vec{0}$. The line therefore has the representation $(x, y, z) = (1, -1, 2) + t(2, 1, -1)$, $t \in \mathbb{R}$. The values $t = 0$ and $t = 1$ give back A and B.

E2. For $A(1, 2, 3)$, the condition $\overrightarrow{AM} \cdot (1, 0, -2) = 0$ is $(x - 1) - 2(z - 3) = 0$. After expanding, the plane has equation $x - 2z + 5 = 0$. The point A satisfies it.

E3. After substitution, $t + 2(1 + t) + (2 - t) = 8$, that is $2t + 4 = 8$, so $t = 2$. The unique intersection is $(2, 3, 0)$; we check $2 + 2 \times 3 + 0 = 8$.

E4. The second equation gives $x = y$. Let $x = y = t$; the first becomes $2t + z = 1$. The set of solutions is $(x, y, z) = (t, t, 1 - 2t) = (0, 0, 1) + t(1, 1, -2)$, a line.

E5. The normals are collinear, the second being twice the first. But doubling $2x - y + z = 3$ gives $4x - 2y + 2z = 6$, which is incompatible with the right-hand side 5. The planes are therefore strictly parallel, not intersecting.

Solutions to Homework 12

1. The coefficients of x, y, z give the non-zero normal $\vec{n} = (1, 2, 2)$ in the orthonormal frame.
2. Look for H on the normal line through M: $H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We check $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. So H is indeed the orthogonal projection, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.
3. The normals $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the planes meet along a line. Since $x = y$, let $x = y = s$. The first equation gives $3s + 2z = 9$, that is $z = \frac{9-3s}{2}$. The intersection is the line $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.
4. The point H has $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it belongs neither to $\mathcal{Q}$ nor to the line of intersection.
5. The cote is the z-coordinate. Setting it to zero requires $9 - 3s = 0$, so $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result is consistent with the property of the orthogonal projection: among all points of the plane, H is the closest to M.

Solutions to the textbook or past-paper work

1. True: the line passes through B and its direction vector $(-2, -1, 3)$ equals $-\frac{1}{2}\overrightarrow{AB}$.
2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.
3. False: for $k = -4$ and $s = \frac{5}{2}$, the two representations give $(11, 12, -14)$. The lines intersect, so they are coplanar.
4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.